

1 - Introduction to Multi-Agent Systems

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Outline

- Introduction
- Syllabus and Course Expectations
- Navigating on Canvas
- Intro to Multi-Agent Systems

About Me (Travis Wiltshire)

- Work
 - Cognition, social and team interaction dynamics
 - Complex systems and coordination between agents
 - Course coordinator for statistics II
- Personal
 - Hiking/Travel
 - [Music](#)
 - Video games



Hiking in Ireland 2025

About Me (Wim Pouw)

- Work
 - Biophysical approach to communication and language
 - Computational tools for quantitative social science
www.envisionbox.org
- Courses
 - Practicum Multi-agent Systems
 - Lecturer Core Topics in Cognitive Science
- Personal
 - Futsal (but sucking at it really badly), boxing
 - Reading and hording (digital) books (cog sci & philosophy)
- Website
 - www.wimpouw.com

Quiz 1

- Where are you from?
- How do you learn?
- What motivates you to participate in class/practicals?
- What do you hope to gain from of this course?
- What do you like to do in your free time?

Course Learning Objectives

- Define and distinguish basic kinds of agents and multi-agent environments;
- Identify fundamental concepts of agent-based modeling and simulation;
- Design agent-based models;
- Evaluate the advantages and challenges related to modeling and simulating multi-agent systems;
- Understand and apply the concepts of verification and validation as applied to agent-based models and simulations;
- Develop and analyze agent-based models to answer research questions about multi-agent systems.

Course Outline (Modules)

- Module 1. Introduction to Multi-Agent Systems
- Module 2. Intro to Agent-Based Modeling
- Module 3. On Using ABMs in Cognitive and Computational Social Science
- Module 4. Exploring and Extending Agent-Based Modeling
- Module 5. Creating Agent-Based Models
- Module 7. Adaptive Agents
- Module 8. Analyzing Agent-Based Models
- Module 9. More on Analysis of ABMs
- Module 10. Example publication combining ABM, MAS, and Cognition
- Module 11. Group Projects

Syllabus and Course Expectations

- The course syllabus is featured on the “Syllabus” tab in Canvas.
- Please read this. Many of your questions may be answered there.

How you should spend your time for this 6EC (~168 hour) course

- Reading 40 hours
- Lectures 21 hours
- Practical Sessions 10.5 hours
- Practice with Netlogo, R, or Python 28 hours
- Project Work 50 hours
- Exams/Prep 18.5 hours

Exams

- A final exam
 - In December
 - Multiple-choice in person and some constructed response
- A resit exam
- All exams are set up by administration not me
 - You must register through Osiris to attend the exams (I can't make exceptions if you miss this)

Group Project

- An agent-based modeling/multi-agent systems research project and report.
- Start thinking about who you'd like to have in your group (and join the groups on Canvas).
- Details and rubric are made available on Canvas.

Practicals (1 hour 45 minutes each)

- Learn how to code up your own data-producing agent-based model
- Focus is on python, and learning by doing (e.g., translating from netlogo to python)
- Practical materials
 - Living quarto document:
https://wimpouw.github.io/multi_agent_systems_practicum/
- Practical 1: Setting up Netlogo and Python models
- Practical 2: Translating a netlogo model into Python
- Practical 2-4: Data-generation, analysis, plotting, complexifying
- Practical 5-6: Group project work with help from Wim

Group Project

- Aim - Use Python or netlogo (and/or any other agent-based modeling program) to develop and experiment with an agent-based model to answer scientific questions about multi-agent systems.
- Free to choose any topic you like as long as you ensure it is connected to cognitive science and artificial intelligence as well as multi-agent systems.
 - team work between humans and autonomous agents (e.g., robots, or virtual)
 - epidemiology (spread of disease in a population)
 - spread of misinformation and factors affecting it
 - cultural transmission of ideas, language, etc.,
 - multi-agent decision making

Requirements for the project report

- Write report of 2,000 (± 200) words
- Problem statement(± 300 words)
- Model description(± 800 words)
- Results(± 600 words)
- Discussion/Conclusions(± 300 words)
- Bibliography – list of references in APA format.
- Appendix – Code for model as a notebook (rmd/jupyter with HTML render)

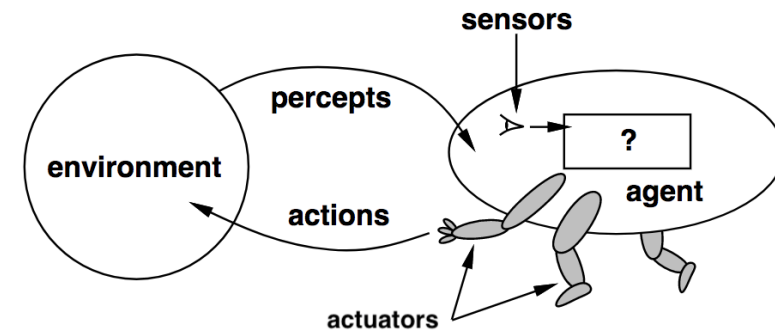
Questions or Comments on Course
Plan/Syllabus?

Emergence Video

- [Emergence - How stupid things become smart together](#)

What is an agent?

- An entity that has perception-action capabilities
 - It can sense its environment and act in it
- Examples of agents
 - Humans, animals, etc
 - Robots, certain kinds of software



Example of an agent

What are Multi-Agent Systems?

- A systems or group of (potentially) interacting agents
 - in some environment that they can sense and act in
 - can communicate and solve problems together
- Can form the bases for distributed AI systems
- We also model multi-agent systems that exist in nature to try to understand how they work (ants, birds, economies, etc.)
- Whole (system) is greater than the sum of the parts (agents)

Some Examples of Multi-Agent Systems

- Cellular/Organs
- Insects Colonies
- Humans
- Organizations/Economies
- Economies
- Video Games
- Autonomous Vehicles
- Drone Swarms

Think-Pair-Share

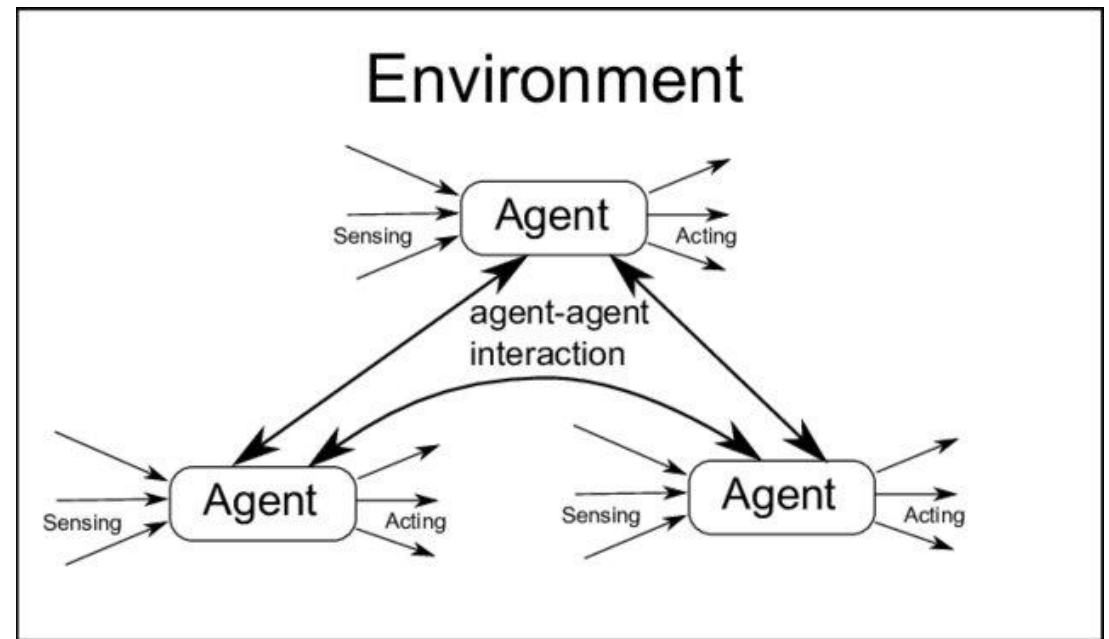
- Think a bit, pair up with a partner, and discuss some ideas for other examples of multi-agent systems.
- Can you think of examples that might appear to be multi-agent systems at first, but maybe don't meet our criteria?

Characteristics of Multi-Agent Systems

- Agent Design
- Environment
- Perception
- Control
- Knowledge
- Communication

Environment

- Static vs dynamic
 - MAS often dynamic (especially with learning and interactions)



Environment (Telgen et al., 2017)

Perception

- Information is distributed in environment
 - Spatially, temporally, semantically
- Partial observability
 - Makes action planning challenging

Control

- Decentralized (emergent, self-organized)
 - Robust
 - Hard to divide decision-making
- Game theory and coordination
- Different control architectures and rules are possible

Knowledge

- Levels of knowledge may differ between agents
- Common or shared knowledge structures are important
 - Knowing what other agents know
 - Shared mental models, situation awareness, transactional memory systems

Communication

- Two-way sender receivers
- Needed for coordination and negotiation
- Protocols for heterogeneous agents

Applications

- E-commerce, trading, auctions
- Robotics
- Computer games
- Social and cognitive science
- Internet
- Human-machine teaming

Challenges

- How can we understand and solve problems with multi-agent systems?
- How can agents maintain a shared understanding of their environment?
- How can we design agents that coordinate and resolve conflicts?
- What kind of learning mechanisms are there for agents?
- How can agents of different types interact effectively?

Questions or Comments?

Preparing for Module 2 - Intro to Agent-Based Modeling

- Complete the readings
 - CH0-CH1, Wilensky & Rand (2015)
- Install Netlogo, Python

Questions?

During this practical session we will do a quiz that covers topics from lectures 5 - 8 and we will work on some Netlogo exercises.